

Stabilisation Policy, Redistribution, And Inequality

Macroeconomics B
Lecture 7 (Chapter 20)

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Where We Are

- ▶ We have derived aggregate demand from IS, the real interest rate, and a Taylor-type policy rule.
- ▶ We have derived aggregate supply from price setting, costs, and expectations.
- ▶ We have combined AS and AD to study demand shocks, supply shocks, and persistence.
- ▶ Today asks a policy question: how should authorities stabilise output and inflation when shocks have unequal effects across households and firms?
- ▶ The lecture moves from positive analysis to normative policy design.

Roadmap For Today

1. Why stabilisation has a distributional motive
2. Rules versus discretion
3. Monetary stabilisation under a Taylor rule
4. Optimal monetary stabilisation
5. Uncertainty, lags, and the lower bound
6. Fiscal policy, automatic stabilisers, and short-time work

What You Will Be Able To Do After This Lecture

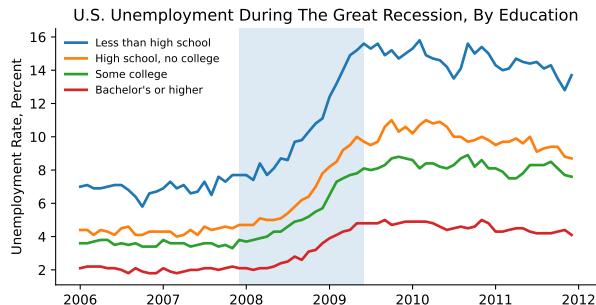
After this lecture, you will be able to:

- ▶ write down and interpret the social loss function.
- ▶ explain why business-cycle stabilisation is also about redistribution and inequality.
- ▶ derive the AD curve implied by a Taylor rule and interpret the roles of h and b .
- ▶ distinguish demand shocks from supply shocks in the stabilisation trade-off.
- ▶ derive the one-period optimal policy curve from iso-loss geometry.
- ▶ explain why fiscal policy and automatic stabilisers become more important when monetary policy is constrained.

Opening Question: Who Pays For A Recession?

- ▶ A recession does not reduce every household's income by the same percentage.
- ▶ Unemployment risk is concentrated among workers with weaker job protection, fewer savings, and lower bargaining power.
- ▶ Inflation also redistributes: borrowers, savers, renters, homeowners, and wage earners are affected differently.
- ▶ A policy that stabilises aggregate output may still shift losses across groups.
- ▶ Today's model gives us a compact way to ask: which aggregate trade-off is chosen, and who is exposed to that choice?

Empirical Fact: Recessions Are Unequal Shocks



Workers with less education saw the largest rise in unemployment during the Great Recession.

Who is most exposed?

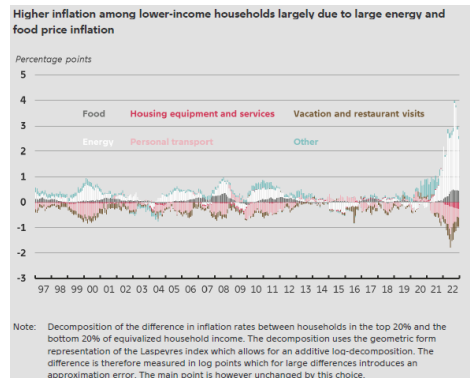
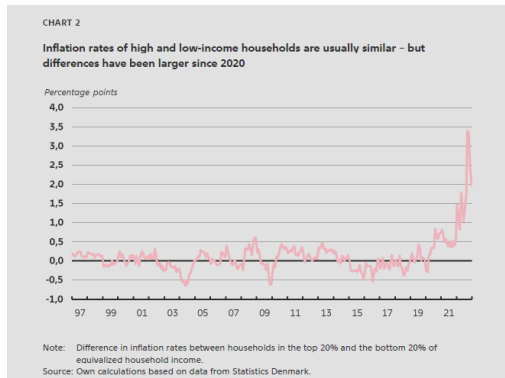
- ▶ Young workers.
- ▶ Low-education workers.
- ▶ Black and Hispanic workers.
- ▶ Men in cyclically sensitive sectors.

$$SL_t = (y_t - \bar{y})^2 + \kappa(\pi_t - \pi^*)^2$$

A lower y_t is not just lower average income. It reallocates risk toward households closer to the unemployment margin.

Sources: Hoynes, Miller and Schaller (2012); BLS unemployment rates by education via FRED.

Empirical Fact: Inflation Is Not The Same Shock For Everyone



Inflation gaps became much larger after the 2020 energy and food price surge.

The difference is largely explained by food, energy, and transport components.

The representative CPI can hide who actually bears the inflation shock: $\pi_t^i = \sum_j \omega_j^i \pi_{j,t}$.

Source: Kuchler, Renkin and Weissert (2023), Danmarks Nationalbank, *Inflation Inequality In Denmark*.

What Is Stabilisation Policy?

Stabilisation policy tries to reduce short-run fluctuations around desired reference points.

- ▶ Output target: keep y_t close to structural output $\bar{y}$.
- ▶ Inflation target: keep π_t close to π^* .
- ▶ Instruments: monetary policy, fiscal policy, and automatic stabilisers.
- ▶ Constraint: policy acts through the AS-AD system, not directly on both variables.
- ▶ Distribution: the remaining losses are not shared equally across households and firms.

The policy problem is to choose feasible points, not ideal points.

The Period Social Loss Function

The period loss is

$$SL_t = (y_t - \bar{y})^2 + \kappa(\pi_t - \pi^*)^2$$

$$SL_t = \hat{y}_t^2 + \kappa \hat{\pi}_t^2, \quad \kappa > 0$$

The target point is

$$(\hat{y}_t, \hat{\pi}_t) = (0, 0).$$

- ▶ $\hat{y}_t = y_t - \bar{y}$: output above or below normal capacity.
- ▶ $\hat{\pi}_t = \pi_t - \pi^*$: inflation above or below target.
- ▶ Squaring means positive and negative gaps both create losses.
- ▶ Squaring also means large gaps are disproportionately costly.
- ▶ κ tells us how much inflation stability matters relative to output stability.

Why Fluctuations Are Costly

The loss function summarises several mechanisms.

- ▶ Risk-averse households prefer stable consumption and employment.
- ▶ Recessions are not shared equally: some workers and firms bear much larger losses.
- ▶ Instability may damage long-run productive capacity through bankruptcies, job loss, and weaker investment.
- ▶ Unpredictable inflation makes long-term contracts and planning harder.
- ▶ A low positive inflation target can reduce the probability of hitting the lower bound.

The model treats these motives as summarised by SL_t .

Forward-Looking Social Loss

A forward-looking authority minimises expected discounted losses:

$$\bar{L}_t = \sum_{i=0}^{\infty} \delta^i \mathbb{E}_t[SL_{t+i}], \quad 0 < \delta < 1$$

- ▶ Current inflation affects future inflation expectations.
- ▶ Current unemployment can create persistence through skill loss, firm exits, and weaker attachment to work.
- ▶ The minimisation is constrained by the economy's structure: demand, supply, and expectations.
- ▶ A policy that looks costly today may prevent larger future losses.
- ▶ Distribution matters because persistent losses often hit the same groups repeatedly.

Marginal Losses And Iso-Loss Slopes

Starting from

$$SL_t = \hat{y}_t^2 + \kappa \hat{\pi}_t^2,$$

the marginal losses are

$$MSL_{y,t} \equiv \frac{\partial SL_t}{\partial \hat{y}_t} = 2\hat{y}_t,$$

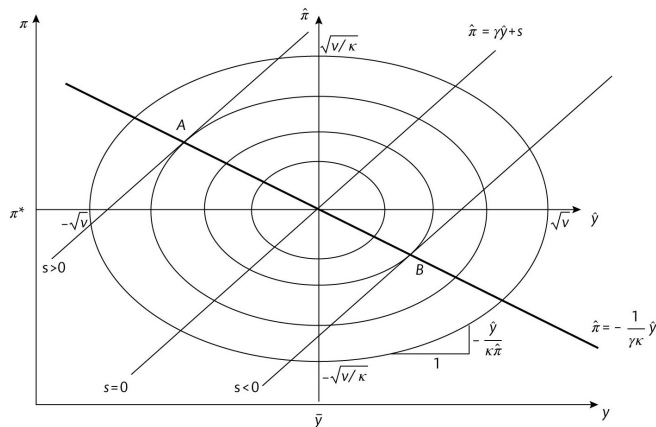
$$MSL_{\pi,t} \equiv \frac{\partial SL_t}{\partial \hat{\pi}_t} = 2\kappa \hat{\pi}_t.$$

- ▶ If $\hat{y}_t < 0$, a higher output gap reduces loss until the target is reached.
- ▶ If $\hat{\pi}_t > 0$, a higher inflation gap raises loss.
- ▶ Absolute marginal losses rise with distance from target.
- ▶ Along a fixed-loss curve:

$$\frac{d\hat{\pi}_t}{d\hat{y}_t} = -\frac{MSL_{y,t}}{MSL_{\pi,t}} = -\frac{\hat{y}_t}{\kappa \hat{\pi}_t}.$$

- ▶ This slope pins down the tangency with the feasible policy trade-off.

Iso-Loss Curves I: Intercepts And Slopes



- Fix one loss level:

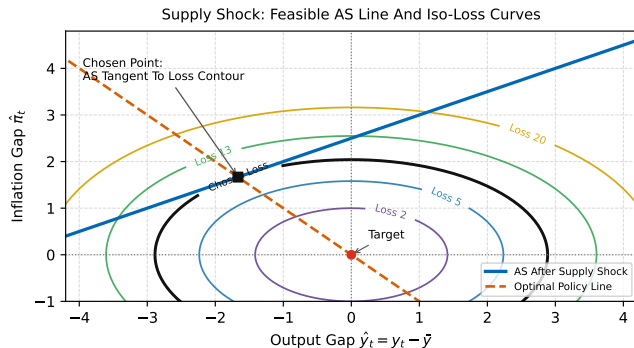
$$\hat{y}_t^2 + \kappa\hat{\pi}_t^2 = v.$$

- Output-axis intercepts: set $\hat{\pi}_t = 0$, so $\hat{y}_t = \pm\sqrt{v}$.
- Inflation-axis intercepts: set $\hat{y}_t = 0$, so $\hat{\pi}_t = \pm\sqrt{v/\kappa}$.
- The slope follows from keeping v fixed:

$$\frac{d\hat{\pi}_t}{d\hat{y}_t} = -\frac{\hat{y}_t}{\kappa\hat{\pi}_t}.$$

- Interpretation: how much inflation-gap movement offsets a one-unit output-gap movement without changing loss.

Iso-Loss Curves II: A Supply-Shock Example



- ▶ The centre is the desired point:
 $\hat{y}_t = \hat{\pi}_t = 0$.
- ▶ An adverse supply shock shifts the feasible AS line up:

$$\hat{\pi}_t = s + \gamma \hat{y}_t, \quad \gamma > 0.$$

- ▶ The authority cannot choose the centre because it is not on the shifted AS line.
- ▶ The best feasible point is where AS touches the lowest reachable loss contour.
- ▶ This tangency gives the policy line:

$$\hat{y}_t = -\gamma \kappa \hat{\pi}_t.$$

▶ Where Does This Come From?

Perspective: From Losses To Policy

- ▶ The objective is clear: minimise expected discounted social loss.
- ▶ The constraint is economic structure: IS demand, AS supply, and expectations.
- ▶ Monetary policy works mainly through the short-term interest rate and the real rate.
- ▶ Fiscal policy works through government demand, taxes, transfers, and automatic stabilisers.
- ▶ The policy question is therefore not only “which instrument moves output?” but “which feasible point gives the lowest loss?”
- ▶ That perspective leads directly to rules versus discretion: systematic responses support expectations, while discretion can target unequal exposure.

Rules Versus Discretion

Policy rule

- ▶ Pre-announced conditional response.
- ▶ Simple enough to be checked.
- ▶ Helps anchor expectations.
- ▶ Example: a Taylor rule.

In practice, credibility requires some predictable reaction pattern.

Discretion

- ▶ Policy chosen case by case.
- ▶ Uses richer information.
- ▶ Flexible in unusual states.
- ▶ May be less credible.

Why Rules Can Help

- ▶ A transparent rule can anchor expectations.
- ▶ If firms and households believe inflation will return to target, wage and price setting become less destabilising.
- ▶ Rule-like fiscal stabilisers respond quickly because they do not require a new political decision.
- ▶ Predictability also matters for distribution: households can plan around known eligibility and replacement rates.

Why Discretion Can Help

- ▶ A simple rule cannot condition on every relevant state of the economy.
- ▶ Deep recessions, financial stress, or sector-specific shocks may require targeted interventions.
- ▶ Discretion can direct support to groups or firms facing unusually large losses.
- ▶ The cost is credibility: repeated discretionary changes may weaken expectations about future policy.

Example: Monetary Policy Rule Versus Discretion

- ▶ Rule: a Taylor-type reaction function raises the policy rate when inflation or output is above target.
- ▶ The rule makes the real-rate response predictable and helps anchor inflation expectations.
- ▶ Discretion: the central bank can depart from the usual reaction when financial stress or the lower bound changes the transmission mechanism.
- ▶ Example: temporary asset purchases, forward guidance, or emergency lending can support demand when the policy rate is constrained.
- ▶ Trade-off: discretion can stabilise unusual states, but repeated departures may weaken credibility.

Example: Fiscal Policy Rule Versus Discretion

- ▶ Rule-like fiscal policy: taxes, transfers, and unemployment insurance react automatically when income and employment fall.
- ▶ Automatic stabilisers are fast and usually well targeted to households hit by recessions.
- ▶ Discretionary fiscal policy: temporary transfers, public investment, or energy support can be scaled to a particular crisis.
- ▶ Trade-off: targeting can improve redistribution, but implementation lags and political design matter.
- ▶ Short-time work is a labour-market example with both rule-like take-up and discretionary crisis expansions.

Denmark: Rule-Like Monetary Policy, Active Fiscal Role

- ▶ Denmark's monetary policy is constrained by the fixed exchange-rate policy vis-à-vis the euro.
- ▶ This limits independent monetary stabilisation through a Danish Taylor rule.
- ▶ Fiscal policy and automatic stabilisers therefore carry more of the domestic stabilisation burden.
- ▶ Redistribution is not a side issue: unemployment benefits, taxes, and transfers shape who absorbs a downturn.
- ▶ Which stabiliser is fastest, and which one is best targeted?

Model Blocks

Aggregate supply

$$\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t.$$

IS curve

$$y_t - \bar{y} = v_t - \alpha_2(r_t - \bar{r}).$$

Policy rule and market rate

$$i_t^p = \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + b(y_t - \bar{y}), \quad i_t = i_t^p + \hat{\rho}_t, \quad r_t = i_t - \pi_{t+1}^e.$$

From The Taylor Rule To AD

Insert the policy rule into IS:

$$y_t - \bar{y} = z_t - a(\pi_t - \pi^*)$$

where

$$z_t \equiv \frac{v_t - \alpha_2 \hat{\rho}_t}{1 + \alpha_2 b}, \quad a \equiv \frac{\alpha_2 h}{1 + \alpha_2 b}.$$

- ▶ b dampens the immediate effect of demand shocks through z_t .
- ▶ h and b jointly determine the AD slope through a .
- ▶ The book's passive central bank has $h = b = 0$.
- ▶ If $h = 0$, AD is vertical in the (y, π) diagram.
- ▶ If $h > 0$, higher inflation triggers higher real rates, lower demand, and a downward-sloping AD curve.

When Is AD Vertical Or Downward Sloping?

The AD curve is

$$\hat{y}_t = z_t - a\hat{\pi}_t, \quad a = \frac{\alpha_2 h}{1 + \alpha_2 b}.$$

- ▶ h is the inflation response in the policy rule.
- ▶ b is the output response in the policy rule.
- ▶ α_2 measures how demand reacts to the real interest rate.

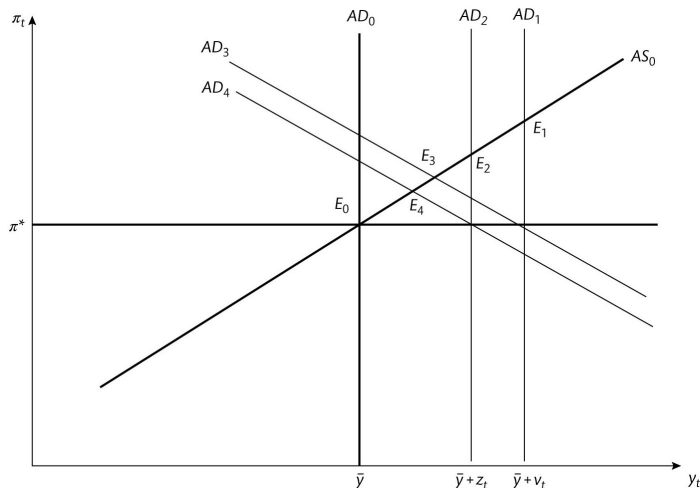
- ▶ **Passive central bank:** $h = b = 0$, so AD is vertical and demand shifts pass through fully.
- ▶ **Vertical AD more generally:** if $h = 0$, then $a = 0$ and $\hat{y}_t = z_t$ for every inflation rate.
- ▶ **Downward-sloping AD:** if $h > 0$, higher inflation raises the real rate, so demand falls.
- ▶ Larger h makes AD flatter in the (y, π) diagram.
- ▶ If $h = 0$, changing b cannot flatten AD because $a = \alpha_2 h / (1 + \alpha_2 b) = 0$.

Demand Shocks: Stabilisation Without A Trade-Off

Assume $s_t = 0$ and a positive demand shock $v_1 > 0$.

- ▶ Passive policy means $h = b = 0$: AD is vertical and the demand shock shifts it right one-for-one.
- ▶ If $b > 0$ but $h = 0$, AD remains vertical: b reduces z_t , but the slope term a is still zero.
- ▶ If $h > 0$, AD slopes downward because higher inflation induces monetary tightening.
- ▶ Combining $h > 0$ and $b > 0$ gives two channels: h creates the slope, while b reduces the shift.
- ▶ Because the economy moves along AS, stabilising output also stabilises inflation on impact.

Demand Shock Diagram



- ▶ AD_1 : passive policy, full rightward shift.
- ▶ AD_2 : output response $b > 0$ shrinks the shift.
- ▶ AD_3 : inflation response $h > 0$ makes AD downward sloping.
- ▶ AD_4 : both responses combine smaller shift and downward slope.
- ▶ With unchanged AS, moving closer to target lowers both gaps.

Demand Shock Cases

For $v_t > 0$ with $s_t = \hat{\rho}_t = 0$:

Case	AD implication	Stabilisation implication
$h = b = 0$	vertical AD, full demand shift	both gaps rise
$h = 0, b > 0$	vertical AD, smaller shift	b dampens the shock
$h > 0, b = 0$	downward-sloping AD	h leans against inflation
$h > 0, b > 0$	downward-sloping AD, smaller shift	both channels work

Large reaction parameters look attractive in this simple demand-shock case, but uncertainty and lags limit that conclusion.

Supply Shocks: Why The Trade-Off Is Real

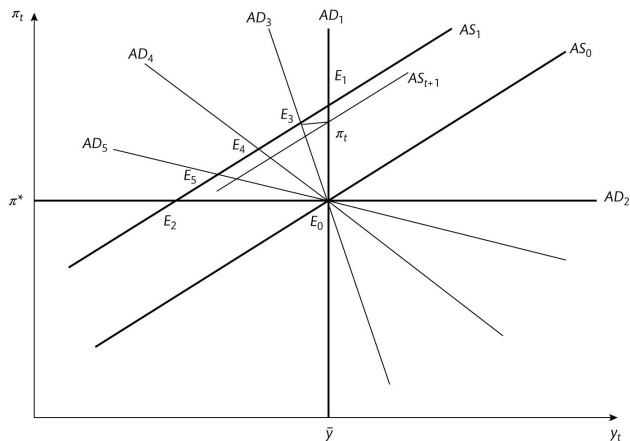
Set $z_t = 0$ and suppose $s_t > 0$:

$$\hat{\pi}_t = \gamma \hat{y}_t + s_t.$$

- ▶ The AS curve shifts upward.
- ▶ At unchanged demand, inflation rises.
- ▶ Lower inflation requires moving down along the new AS curve.
- ▶ That means a lower output gap.

After a supply shock, one instrument cannot stabilise both targets.

Supply Shock Diagram



- ▶ AD_1 : vertical AD; output is stabilised, inflation absorbs the shock.
- ▶ AD_2 : horizontal limit; inflation is stabilised, output falls strongly.
- ▶ AD_3 to AD_5 : choices of h and b select intermediate equilibria.
- ▶ Flatter AD means stronger inflation response h and less inflation accommodation.
- ▶ Since outcomes lie on shifted AS, less inflation means more output loss.

Supply Shock Cases

For $s_t > 0$ with $v_t = \hat{\rho}_t = 0$:

- ▶ If $h = 0$, AD is vertical: output is insulated on impact, and inflation absorbs the shock.
- ▶ If $h > 0$, AD slopes downward: higher inflation triggers tighter policy and lower demand.
- ▶ If h is very large while b is finite, $a \rightarrow \infty$ and AD approaches a horizontal line.
- ▶ The value of b affects a , but h is the direct way to choose the AD slope for a given b .
- ▶ Appropriate choices of h and b select equilibria between the extremes, such as E_3 to E_5 .
- ▶ Distributional angle: energy and food shocks hit low-income households directly, while disinflation raises unemployment risk.

Supply shocks turn stabilisation into a choice of relative losses.

Socratic Group Exercise: Short Text Only

Go to gosocrative.com and enter room MACROECONOMICSB. Work in small groups and enter short text answers.

1. Suppose $s_t > 0$ and society increases κ . What should happen to the chosen output gap? Explain the mechanism.
2. For Denmark, would broad fiscal stimulus be a good response to an imported energy shock? Give one model-based reason.

Visit gosocrative.com and enter room name MACROECONOMICSB



Optimal Monetary Policy Without A Taylor Rule

To understand optimal monetary policy, first remove the operational Taylor rule.

$$y_t - \bar{y} = v_t - \alpha_2(r_t - \bar{r}), \quad \text{IS}$$

$$r_t = r_t^p + \hat{\rho}_t, \quad \text{RP}$$

$$\pi_t = \pi_{t-1} + \gamma(y_t - \bar{y}) + s_t. \quad \text{AS}$$

- ▶ The central bank chooses the policy rate i_t^p and therefore the policy real rate

$$r_t^p \equiv i_t^p - \pi_{t+1}^e.$$

- ▶ Through IS, choosing r_t^p indirectly chooses aggregate demand.
- ▶ Static expectations imply $\pi_t^e = \pi_{t-1}$ in the AS curve.

The monetary-policy choice is not constrained by IS itself. The binding trade-off comes from AS.

The Central Bank Chooses A Feasible Point On AS

Think of policy as indirectly choosing $\hat{y}_t \equiv y_t - \bar{y}$.

$$\hat{\pi}_t = \gamma \hat{y}_t + s_t \quad \text{if } \pi_t^e = \pi^*.$$

- ▶ A demand shock can be offset by choosing the real rate appropriately.
- ▶ A supply shock shifts the AS constraint itself.
- ▶ The central bank chooses the point on the shifted AS curve with the lowest social loss.

$$\min_{\hat{y}_t} \hat{y}_t^2 + \kappa(\gamma \hat{y}_t + s_t)^2.$$

The Optimal Policy Problem

For a one-period problem with $\pi_t^e = \pi^*$ and $z_t = 0$:

$$\begin{array}{ll} \min_{\hat{y}_t} & SL_t = \hat{y}_t^2 + \kappa \hat{\pi}_t^2, \\ \text{s.t.} & \hat{\pi}_t = \gamma \hat{y}_t + s_t. \end{array}$$

where $\hat{y}_t \equiv y_t - \bar{y}$ and $\hat{\pi}_t \equiv \pi_t - \pi^*$.

The first-order condition implies the optimal policy line:

$$\hat{\pi}_t = -\frac{1}{\gamma\kappa} \hat{y}_t.$$

Intuition

- ▶ The real rate lets the bank choose aggregate demand.
- ▶ AS then maps this choice into inflation.
- ▶ A supply shock makes the target point infeasible.
- ▶ Lower inflation requires accepting a lower output gap.

Derivation

Optimal Responses To An Adverse Supply Shock

Combine AS with the optimal policy line:

$$\hat{\pi}_t = \gamma \hat{y}_t + s_t, \quad \hat{\pi}_t = -\frac{1}{\gamma \kappa} \hat{y}_t.$$

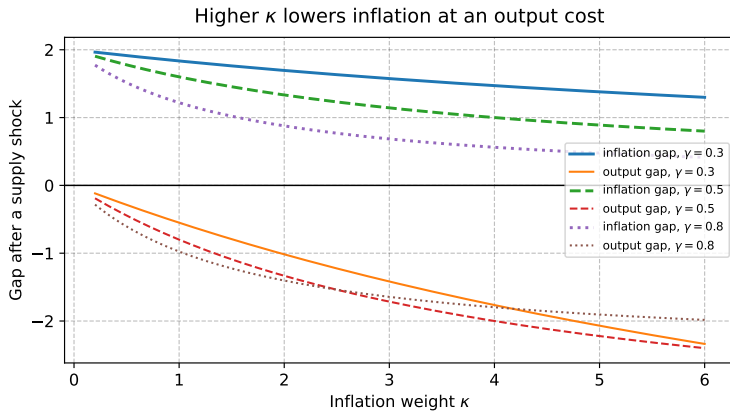
Solving gives

$$\hat{y}_t^* = -\frac{\gamma \kappa}{1 + \gamma^2 \kappa} s_t, \quad \hat{\pi}_t^* = \frac{1}{1 + \gamma^2 \kappa} s_t.$$

- ▶ Policy accepts some inflation because fully stabilising inflation would require too large an output fall.
- ▶ Policy accepts some output loss because fully accommodating inflation would be too costly.

▶ Intermediate Steps

What Changes With κ And γ ?



Self-generated model figure. Each line uses the optimal response formulas for a fixed adverse supply shock $s = 2$. A higher inflation weight lowers the inflation gap but makes the output gap more negative.

Optimal Policy And The Taylor Rule

Under a Taylor rule the AD curve is

$$\hat{y}_t = z_t - a(\pi_t - \pi^*), \quad a = \frac{\alpha_2 h}{1 + \alpha_2 b}.$$

The optimal policy line after a supply shock is

$$\hat{y}_t = -\gamma\kappa(\pi_t - \pi^*).$$

Thus a Taylor rule can implement the one-period optimum if

$$a = \gamma\kappa.$$

- ▶ Use b to dampen demand shocks.
- ▶ Use h to set the trade-off after supply shocks.

Forward-Looking Monetary Policy

The one-period optimum is useful, but it misses a dynamic channel.

- ▶ The bank can choose a point on the current AS curve through the policy real rate.
- ▶ It chooses the current inflation gap and plans future inflation gaps.
- ▶ Future shocks have expected value zero, but current inflation becomes next period's expected inflation.
- ▶ Higher inflation today therefore shifts next period's AS curve upward.

$$\hat{y}_t = -\gamma\kappa\hat{\pi}_t + \delta E_t(\hat{y}_{t+1})$$

▶ Dynamic Derivation

Forward-Looking Policy Line

The forward-looking condition can be summarised by a policy line

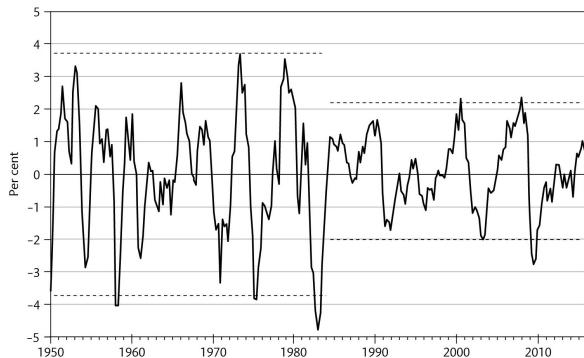
$$\hat{\pi}_t = -\varphi \hat{y}_t, \quad 0 < \varphi < \frac{1}{\gamma\kappa}.$$

Quick Intuition

- ▶ The myopic bank only weighs today's output loss against today's inflation loss.
- ▶ A forward-looking bank also sees that inflation today becomes expected inflation tomorrow.
- ▶ Current inflation therefore worsens the next AS trade-off.
- ▶ The policy line becomes flatter: for a given output loss, policy allows less inflation.

▶ Policy-Line Algebra

The Great Moderation: Context



US real GDP cyclical component, 1950Q1–2015Q4. Dashed bands show 1.96 standard deviations before and after 1984.

- ▶ The “Great Moderation” refers to lower output and inflation volatility from the mid-1980s until the financial crisis.
- ▶ One interpretation: monetary policy became more active after the Volcker disinflation.
- ▶ In Taylor-rule language, the central bank reacted more strongly to inflation and the output gap.
- ▶ Other explanations also matter: smaller shocks, structural change, better inventories, and luck.
- ▶ The lesson for this lecture: rules can stabilise, but they do not remove crises or distributional costs.

Why Very Strong Rules Can Be Risky

- ▶ GDP is measured with delay and later revised.
- ▶ The output gap is not observed directly.
- ▶ Inflation measures may not match the cost pressures relevant for every household.
- ▶ Credit spreads and risk premia can move independently of the policy rate.
- ▶ If h and b are too large, policy may react strongly to measurement noise.

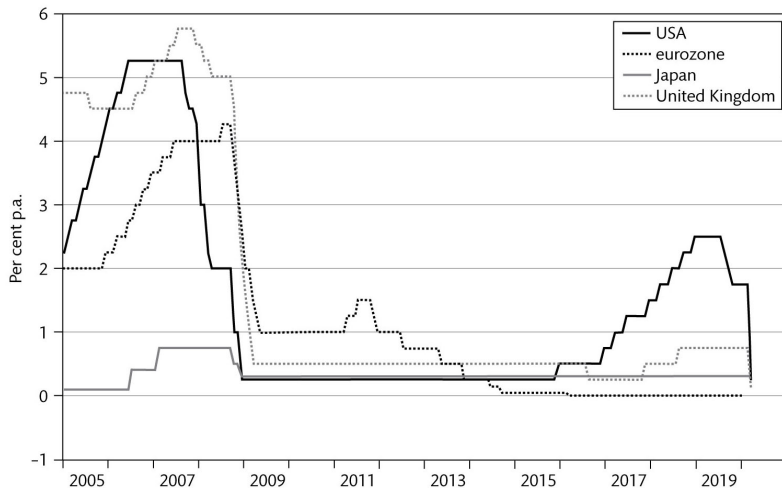
Inside And Outside Lags

- ▶ **Inside lag:** recognition, decision, and implementation.
- ▶ **Outside lag:** time from implementation to peak effect.
- ▶ Monetary policy usually has a shorter inside lag but a longer outside lag.
- ▶ Fiscal policy often has a longer inside lag, unless it is automatic.
- ▶ Automatic stabilisers are valuable because they combine speed with redistribution.

Automatic Stabilisers As Insurance

- ▶ Taxes and transfers reduce the fall in disposable income when market income drops.
- ▶ Unemployment insurance targets households with direct income loss.
- ▶ Progressive taxes reduce disposable-income volatility at the top and bottom of the cycle.
- ▶ This can stabilise aggregate demand because liquidity-constrained households have high marginal propensities to consume.
- ▶ The macro effect and the inequality effect point in the same direction.

The Lower Bound Constraint



When nominal policy rates approach the lower bound, conventional monetary stabilisation weakens.

Liquidity Trap Model

IS with a fiscal impulse Δ_t :

$$y_t - \bar{y} = \Delta_t - \alpha_2(i_t - \pi_{t+1}^e - \bar{r}) + v_t.$$

Policy with a lower bound:

$$i_t = \begin{cases} \bar{r} + \pi_t + h(\pi_t - \pi^*) + b(y_t - \bar{y}), & \text{if this is non-negative,} \\ 0, & \text{otherwise.} \end{cases}$$

AS remains

$$\pi_t = \pi_t^e + \gamma(y_t - \bar{y}) + s_t.$$

Derivation

The LT Line

The LT line is the set of (y_t, π_t) combinations where the Taylor rule implies $i_t = 0$:

$$0 = \bar{r} + \pi_t + h(\pi_t - \pi^*) + b(y_t - \bar{y}).$$

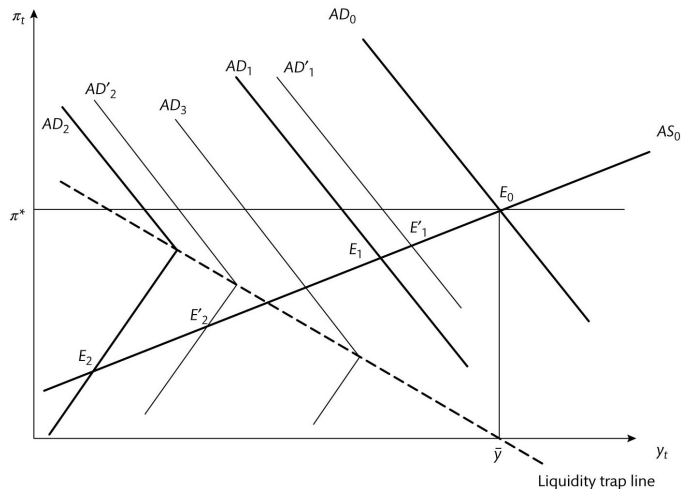
Solving for the output gap:

$$y_t - \bar{y} = \frac{h}{b}\pi^* - \frac{\bar{r}}{b} - \left(1 + \frac{h}{b}\right)\pi_t$$

- ▶ Above the LT line, normal Taylor-rule AD applies.
- ▶ Below the LT line, the nominal rate is stuck at the lower bound.

Derivation

Liquidity Trap Diagram



Below the LT line, higher inflation lowers the real rate and can make AD upward sloping.

- ▶ E_0 is the normal equilibrium before the shock.
- ▶ Falling demand shifts AD left.
- ▶ Above the LT line, policy can cut the nominal rate.
- ▶ At the lower bound, the nominal rate cannot fall further.
- ▶ Higher inflation lowers the real rate and supports demand.
- ▶ AD can bend upward inside the trap.

Why Fiscal Multipliers Rise At The Lower Bound

Δ_t is the fiscal demand impulse: extra government demand or transfers that raise spending.

Outside the liquidity trap:

$$y_t - \bar{y} = \frac{v_t + \Delta_t}{1 + \alpha_2 b} - \frac{\alpha_2 h}{1 + \alpha_2 b} (\pi_t - \pi^*).$$

- ▶ The Taylor rule raises rates when demand or inflation rises.
- ▶ Only part of Δ_t reaches output; the rest is crowded out.

Inside the liquidity trap:

$$y_t - \bar{y} = \Delta_t + \alpha_2 (\pi_t + \bar{r}) + v_t.$$

- ▶ The nominal rate is stuck, so there is no offsetting rate hike.
- ▶ If inflation rises, the real rate falls and demand is reinforced.

Derivation

Fiscal Policy, Redistribution, And Inequality

- ▶ Fiscal policy can stabilise aggregate demand and redistribute losses at the same time.
- ▶ Transfers to liquidity-constrained households have a stronger demand effect than transfers to unconstrained households.
- ▶ Public employment support can prevent persistent job loss and skill depreciation.
- ▶ Poorly targeted support can preserve low-productivity matches or fuel inflation.
- ▶ The rules-versus-discretion question is therefore also a targeting question.

Back To The Opening Question

- ▶ The aggregate model says demand shocks can be stabilised without an inflation-output trade-off on impact.
- ▶ Supply shocks force a choice between inflation and output stabilisation.
- ▶ That choice is distributional because inflation and unemployment risk are unequally allocated.
- ▶ Rules support credibility and speed, while discretion supports targeting in unusual states.
- ▶ Good stabilisation policy asks both: what happens to aggregate loss, and who bears the remaining loss?

Takeaways

- ▶ This lecture turns the AS-AD model into a policy-design problem.
- ▶ The social loss function summarises output-gap and inflation-gap costs.
- ▶ Redistribution matters because recessions and inflation do not affect all households equally.
- ▶ Demand shocks can be stabilised more easily than supply shocks.
- ▶ A Taylor rule can implement the optimal policy curve under strong assumptions.
- ▶ Data uncertainty and lags argue against mechanically extreme rule parameters.
- ▶ At the lower bound, fiscal policy and automatic stabilisers become more powerful.
- ▶ Short-time work illustrates why discretion can be valuable in deep crises but problematic in expansions.

Preview: Next Lecture

- ▶ We continue with fiscal stabilisation in more detail.
- ▶ The focus shifts from monetary policy to government spending, taxes, and debt.
- ▶ We will ask how automatic stabilisers and discretionary fiscal packages affect output and inflation.
- ▶ The exam skill is to connect a policy instrument to the relevant constraint, shock, and welfare trade-off.

Appendix: Socratic Short-Text Answers

1. If $s_t > 0$ and society increases κ , the chosen output gap should become more negative.
 - ▶ A higher κ means inflation deviations are more costly relative to output deviations.
 - ▶ The policy line $\hat{y}_t = -\gamma\kappa\hat{\pi}_t$ becomes steeper in the $(\hat{\pi}_t, \hat{y}_t)$ trade-off.
 - ▶ Inflation is reduced more aggressively, so policy accepts a larger recessionary output gap.
2. Broad fiscal stimulus is usually a poor response to an imported energy shock.
 - ▶ The shock is primarily a supply shock, so extra aggregate demand raises inflation pressure.
 - ▶ Better fiscal policy is targeted insurance: temporary transfers or energy support for exposed households.
 - ▶ Broad stimulus may protect average income but worsens the inflation-output trade-off.

Appendix: AD Curve Under A Taylor Rule

Start from IS and the real rate:

$$y_t - \bar{y} = v_t - \alpha_2(r_t - \bar{r}), \quad r_t = i_t - \pi_{t+1}^e.$$

The Taylor rule and spread shock imply

$$i_t = \bar{r} + \pi_{t+1}^e + h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t.$$

Therefore

$$r_t - \bar{r} = h(\pi_t - \pi^*) + b(y_t - \bar{y}) + \hat{\rho}_t.$$

Insert into IS and solve:

$$y_t - \bar{y} = \frac{v_t - \alpha_2 \hat{\rho}_t}{1 + \alpha_2 b} - \frac{\alpha_2 h}{1 + \alpha_2 b} (\pi_t - \pi^*).$$

Appendix: One-Period Optimal Policy Setup

With the Taylor rule removed, the model is

$$\hat{y}_t = v_t - \alpha_2(r_t - \bar{r}),$$

$$r_t = r_t^p + \hat{\rho}_t,$$

$$\pi_t = \pi_{t-1} + \gamma \hat{y}_t + s_t.$$

If the bank disregards future periods, it minimises current-period loss:

$$SL_t = \hat{y}_t^2 + \kappa \hat{\pi}_t^2.$$

For the one-period supply-shock case, use $\pi_t^e = \pi^*$ and $z_t = 0$:

$$\hat{\pi}_t = \gamma \hat{y}_t + s_t.$$

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Appendix: Where The Policy Line Comes From

The central bank chooses a feasible point on AS:

$$\widehat{\pi}_t = s_t + \gamma \widehat{y}_t.$$

Along this constraint,

$$\frac{d\widehat{\pi}_t}{d\widehat{y}_t} = \gamma.$$

The loss function is

$$SL_t = \widehat{y}_t^2 + \kappa \widehat{\pi}_t^2.$$

A fixed-loss contour satisfies

$$dSL_t = 2\widehat{y}_t d\widehat{y}_t + 2\kappa \widehat{\pi}_t d\widehat{\pi}_t = 0.$$

Therefore the iso-loss slope is

$$\left. \frac{d\widehat{\pi}_t}{d\widehat{y}_t} \right|_{SL} = -\frac{\widehat{y}_t}{\kappa \widehat{\pi}_t}.$$

At the best feasible point, AS is tangent to the lowest reachable loss contour:

$$-\frac{\widehat{y}_t}{\kappa \widehat{\pi}_t} = \gamma.$$

Rearranging gives the policy line:

$$\boxed{\widehat{y}_t = -\gamma \kappa \widehat{\pi}_t} \quad \Leftrightarrow \quad \boxed{\widehat{\pi}_t = -\frac{1}{\gamma \kappa} \widehat{y}_t}.$$

This is not a new structural equation. It is the optimality condition balancing output-gap loss against inflation-gap loss along AS.

Appendix: Solving For The Inflation Response

Combine the AS constraint and the optimal policy line:

$$\hat{\pi}_t = \gamma \hat{y}_t + s_t, \quad \hat{\pi}_t = -\frac{1}{\gamma \kappa} \hat{y}_t.$$

First rewrite the policy line as

$$\hat{y}_t = -\gamma \kappa \hat{\pi}_t.$$

Substitute this into AS:

$$\hat{\pi}_t = \gamma(-\gamma \kappa \hat{\pi}_t) + s_t = -\gamma^2 \kappa \hat{\pi}_t + s_t.$$

Move the inflation-gap terms to the left:

$$(1 + \gamma^2 \kappa) \hat{\pi}_t = s_t \quad \Rightarrow \quad \boxed{\hat{\pi}_t^* = \frac{1}{1 + \gamma^2 \kappa} s_t.}$$

Appendix: Solving For The Output Response

Use the optimal policy line again:

$$\hat{y}_t = -\gamma\kappa\hat{\pi}_t.$$

Insert the inflation response from the previous slide:

$$\hat{y}_t^* = -\gamma\kappa \left(\frac{1}{1 + \gamma^2\kappa} s_t \right) = \boxed{-\frac{\gamma\kappa}{1 + \gamma^2\kappa} s_t}.$$

Check the solution in AS:

$$\gamma\hat{y}_t^* + s_t = -\frac{\gamma^2\kappa}{1 + \gamma^2\kappa} s_t + s_t = \frac{1}{1 + \gamma^2\kappa} s_t = \hat{\pi}_t^*.$$

◀ Inflation Gap

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▶ Myopic Case

Appendix: If The Central Bank Disregards The Future

The one-period problem is the limiting case of the forward-looking objective when $\delta \rightarrow 0$:

$$\bar{L}_t = \sum_{i=0}^{\infty} \delta^i E_t(SL_{t+i}) \longrightarrow SL_t.$$

- ▶ The bank treats realised shocks in period t as the only welfare-relevant problem.
- ▶ It ignores that inflation today can shift expected inflation and next period's AS curve.
- ▶ This makes current inflation deviations look less costly than under a forward-looking objective.
- ▶ In the richer dynamic problem, persistence usually calls for a stronger inflation response.

◀ Responses

▶ Forward-Looking Case

Main Slide

Appendix: Forward-Looking Setup

The bank can move the current economy along AS by choosing the policy real rate. It is convenient to treat $\hat{\pi}_t$ as the choice variable:

$$\hat{y}_t = \frac{\hat{\pi}_t - \hat{\pi}_{t-1}}{\gamma} - \frac{s_t}{\gamma}.$$

Rolling AS forward gives

$$\hat{y}_{t+\tau} = \frac{\hat{\pi}_{t+\tau} - \hat{\pi}_{t+\tau-1}}{\gamma} - \frac{s_{t+\tau}}{\gamma}, \quad \tau \geq 1.$$

- ▶ Current s_t is realised, so the current output gap is pinned down once $\hat{\pi}_t$ is chosen.
- ▶ Future shocks have expected value zero: $E_t(s_{t+\tau}) = 0$.
- ▶ The bank plans $\hat{\pi}_{t+\tau}$ for $\tau \geq 0$, implements $\hat{\pi}_t$, and re-optimises next period.

Appendix: Forward-Looking Loss Terms

The objective is the expected discounted sum of period losses:

$$\bar{L}_t = \hat{y}_t^2 + \kappa \hat{\pi}_t^2 + \delta E_t(\hat{y}_{t+1}^2) + \delta \kappa \hat{\pi}_{t+1}^2 + \delta^2 E_t(\hat{y}_{t+2}^2) + \delta^2 \kappa \hat{\pi}_{t+2}^2 + \dots$$

Using the AS expressions:

$$\begin{aligned} \bar{L}_t = & \left(\frac{\hat{\pi}_t - \hat{\pi}_{t-1}}{\gamma} - \frac{s_t}{\gamma} \right)^2 + \kappa \hat{\pi}_t^2 \\ & + \delta E_t \left[\left(\frac{\hat{\pi}_{t+1} - \hat{\pi}_t}{\gamma} - \frac{s_{t+1}}{\gamma} \right)^2 \right] + \delta \kappa \hat{\pi}_{t+1}^2 + \dots \end{aligned}$$

◀ Setup

▶ FOC

Appendix: Forward-Looking First-Order Condition

Only the first three terms matter for the derivative with respect to $\hat{\pi}_t$:

$$\frac{\partial \bar{L}_t}{\partial \hat{\pi}_t} = \frac{2}{\gamma} \hat{y}_t + 2\kappa \hat{\pi}_t - \frac{2\delta}{\gamma} E_t(\hat{y}_{t+1}) = 0.$$

Multiplying by γ gives

$$2\hat{y}_t + 2\gamma\kappa\hat{\pi}_t - 2\delta E_t(\hat{y}_{t+1}) = 0.$$

Therefore

$$\boxed{\hat{y}_t = -\gamma\kappa\hat{\pi}_t + \delta E_t(\hat{y}_{t+1})} \Leftrightarrow \boxed{\hat{\pi}_t = -\frac{1}{\gamma\kappa}\hat{y}_t + \frac{\delta}{\gamma\kappa}E_t(\hat{y}_{t+1})}.$$

For $\delta = 0$, this collapses to the myopic policy line.

◀ Loss Terms

▶ Policy Line

Appendix: Forward-Looking Policy Line

Try to enforce a linear policy line in every period:

$$\hat{\pi}_t = -\varphi \hat{y}_t, \quad \varphi > 0.$$

For period $t + 1$, combine this policy line with AS:

$$\hat{\pi}_{t+1} = \hat{\pi}_t + \gamma \hat{y}_{t+1} + s_{t+1}, \quad \hat{\pi}_{t+1} = -\varphi \hat{y}_{t+1}.$$

Then

$$E_t(\hat{y}_{t+1}) = -\frac{\hat{\pi}_t}{\gamma + \varphi}.$$

Insert this into the forward-looking FOC:

$$\hat{\pi}_t = -\frac{1}{\gamma \kappa} \hat{y}_t - \frac{\delta}{\gamma \kappa (\gamma + \varphi)} \hat{\pi}_t.$$

The policy-line slope solves

$$\varphi = \frac{1}{\gamma \kappa + \frac{\delta}{\gamma + \varphi}}.$$

Appendix: Implementing The Forward-Looking Line

The Taylor rule creates the AD curve

$$\hat{y}_t = z_t - a\hat{\pi}_t, \quad a = \frac{\alpha_2 h}{1 + \alpha_2 b}.$$

To enforce $\hat{\pi}_t = -\varphi\hat{y}_t$, rewrite the policy line as

$$\hat{y}_t = -\frac{1}{\varphi}\hat{\pi}_t.$$

Thus the AD slope should satisfy

$$a = \frac{1}{\varphi}.$$

- ▶ Larger b reduces the horizontal demand-shock term z_t .
- ▶ Positive h is the Taylor-principle response to persistent inflation.
- ▶ With uncertainty and lags, extreme h and b are risky even if the algebra points to strong responses.

Appendix: Liquidity Trap Model Equations

Start from the IS curve with government demand, taxes, and other demand shocks collected in one fiscal impulse:

$$y_t - \bar{y} = \Delta_t - \alpha_2(r_t - \bar{r}) + v_t.$$

Use the ex ante real rate:

$$r_t = i_t - \pi_{t+1}^e.$$

Substitution gives

$$y_t - \bar{y} = \Delta_t - \alpha_2(i_t - \pi_{t+1}^e - \bar{r}) + v_t.$$

The lower bound modifies the policy rule:

$$i_t = \max\{0, \bar{r} + \pi_t + h(\pi_t - \pi^*) + b(y_t - \bar{y})\}.$$

Appendix: Deriving The LT Line

At the LT line, the unconstrained Taylor rule implies exactly zero:

$$0 = \bar{r} + \pi_t + h(\pi_t - \pi^*) + b(y_t - \bar{y}).$$

Expand the inflation term:

$$0 = \bar{r} + \pi_t + h\pi_t - h\pi^* + b(y_t - \bar{y}).$$

Collect and solve for the output gap:

$$b(y_t - \bar{y}) = h\pi^* - \bar{r} - (1 + h)\pi_t.$$

$$y_t - \bar{y} = \frac{h}{b}\pi^* - \frac{\bar{r}}{b} - \left(1 + \frac{h}{b}\right)\pi_t$$

Appendix: Fiscal Multipliers At The Lower Bound

Outside the liquidity trap, insert the Taylor rule into IS:

$$y_t - \bar{y} = \Delta_t + v_t - \alpha_2 [h(\pi_t - \pi^*) + b(y_t - \bar{y})].$$

Collect output terms:

$$(1 + \alpha_2 b)(y_t - \bar{y}) = v_t + \Delta_t - \alpha_2 h(\pi_t - \pi^*).$$

$$y_t - \bar{y} = \frac{v_t + \Delta_t}{1 + \alpha_2 b} - \frac{\alpha_2 h}{1 + \alpha_2 b}(\pi_t - \pi^*)$$

Inside the liquidity trap, $i_t = 0$ and $\pi_{t+1}^e = \pi_t$:

$$y_t - \bar{y} = \Delta_t + \alpha_2(\pi_t + \bar{r}) + v_t$$

Appendix: Short-Time Work As Rule And Discretion

- ▶ Short-time work subsidises temporary hours reductions so firms can retain workers instead of laying them off.
- ▶ Rule-like element: take-up rises automatically when demand falls and firms reduce hours.
- ▶ Discretionary element: governments can expand eligibility, duration, or generosity in deep crises.
- ▶ Redistribution channel: the policy protects employment relationships for workers exposed to layoff risk.
- ▶ Policy risk: the same instrument can be valuable in recessions and wasteful in expansions.

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Evidence

Appendix: Short-Time Work Evidence

Gehrke and Hochmuth study German short-time work with a smooth-transition VAR.

- ▶ Discretionary STW policy has much stronger employment effects in recessions than in expansions.
- ▶ In the Great Recession, the maximum effect is estimated at up to 0.87 jobs per discretionary short-time worker.
- ▶ The same type of policy can have small or even negative employment effects in expansions.
- ▶ Interpretation: targeted discretion is most defensible when the state of the economy makes job destruction especially costly.

Gehrke and Hochmuth, *Scandinavian Journal of Economics*, 2021.

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STW Appendix

Appendix: Distributional Loss

One way to represent inequality concerns is to add labour-market exposure:

$$SL_t^D = \hat{y}_t^2 + \kappa \hat{\pi}_t^2 + \lambda U_t^{\text{exposed}}, \quad \lambda > 0.$$

- ▶ U_t^{exposed} captures unemployment risk among low-liquidity workers.
- ▶ This is not the baseline model.
- ▶ It shows why equal aggregate gaps can imply different welfare losses.
- ▶ The notebook uses this only as a numerical exercise.